FPT Education

FPT UNIVERSITY

SOFTWARE TESTING (SWT301)

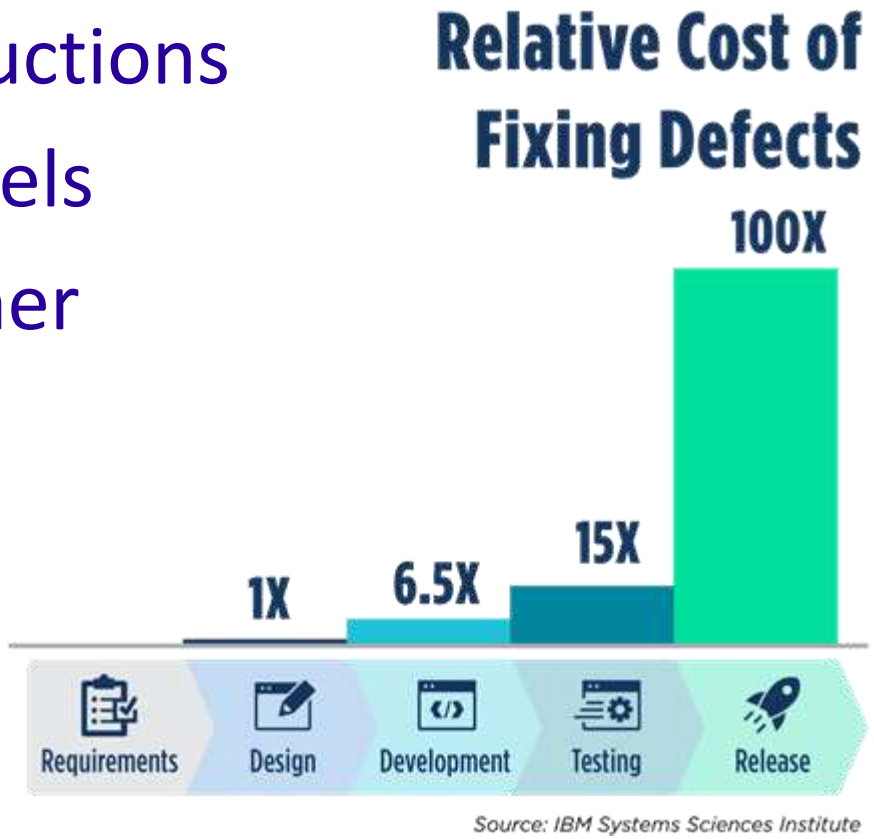
STATIC TECHNIQUES (CHAPTER 3)

Static Testing

- Static testing techniques are those techniques that test software and work-products without executing the code
- Static techniques find the causes of failures rather than the failure itself, which would be found during test execution
- Reviews are normally completed manually; static analysis is normally completed automatically using tools

Static Testing Benefits

- Development productivity improvement
- Reduced development timescales
- Reduced testing time and cost
- Lifetime cost reductions
- Reduced fault levels
- Improved customer relations
- etc.



Static Testing

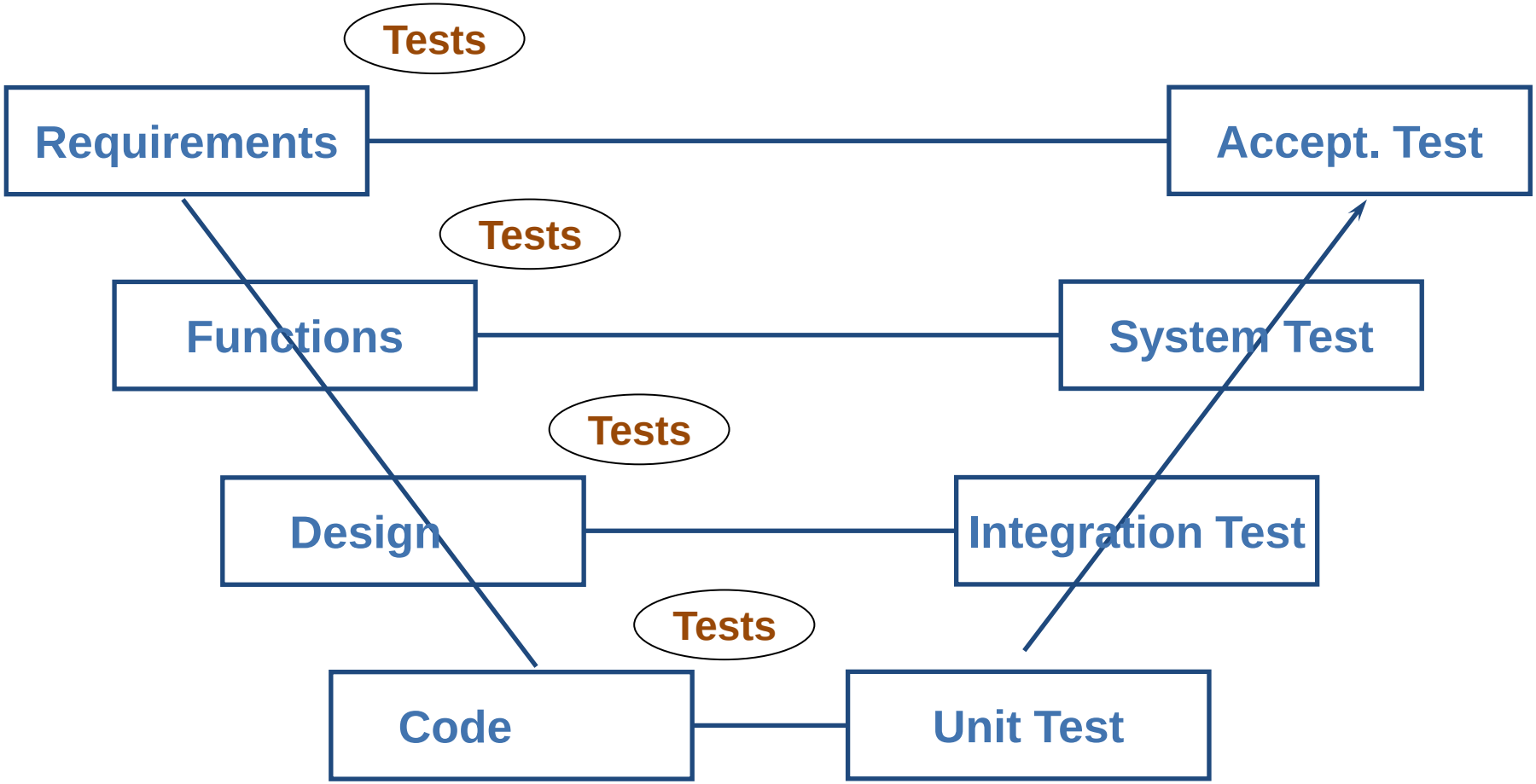
What can be inspected?

- policy, strategy, business plans, marketing or advertising material, contracts
- system requirements, feasibility studies, acceptance test plans
- test plans, test designs, test cases, test results
- system designs, logical & physical
- software code
- user manuals, procedures, training material

Anything written down can be Inspected

Static Testing

What can be inspected?



Static Testing
People Techniques

- individual:
  - desk-checking, data-stepping, proof-reading
- group:
  - Reviews (informal & formal): for consensus
  - Walkthrough: for education
  - Inspection (most formal): to find faults

Static techniques do not execute code

Static Testing

The types of defects found by reviews

- Requirement defects (e.g., inconsistencies, ambiguities, contradictions)
- Design defects (inefficient algorithms, DB structure,..)
- Coding defects (e.g., unreachable code, duplicate code)
- Deviations from standards (lack of adherence to coding standard)
- Incorrect interface specifications
- Security vulnerabilities (e.g., buffer overflow)
- Traceability problems (gaps / inaccuracies / lack of coverage)
- Maintainability defects (e.g., poor reusability, code smells)

Static Testing
Other Objectives

- Educational
- Mutual understandings
- Decision-making facilitation
- Agreed commitment

Review Process 1/7

Review Process

- Planning
- Initiate review
- Individual review
- Issue comms & analysis
- Fixing & reporting

Roles & Responsibilities

- Author
- Management
- Facilitator
- Review leader
- Reviewer
- Scribe

Review Types

- Informal review
- Walkthrough
- Technical review
- Inspection

Review Techniques

- Ad hoc
- Checklist-based
- Scenario-based & Dry runs
- Role-based
- Perspective-based

Review Process 2/7

Reviews

- Review – A type of static testing during which a work product or process is evaluated by one/more individuals to detect issues & to provide improvements.
  - Formality level depends on factors such as maturity of dev. process, legal/regulatory reqs., need for audit trail.
- Informal review – A type of review without a formal (documented) procedure.
  - most common type of review
- Formal review – A form of review that follows a defined process with a formally documented output.

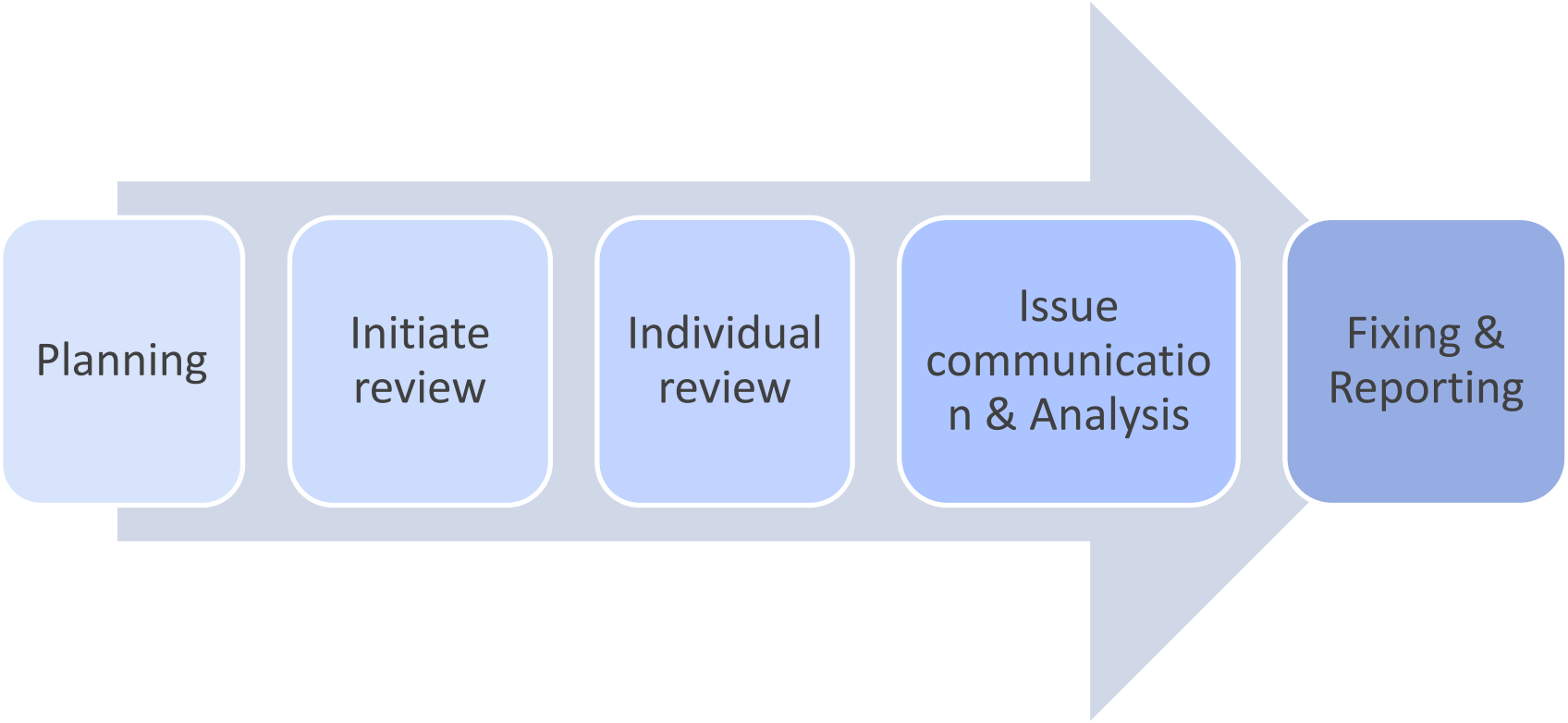
Review Process 3/7

Basic review process

- The document under review is studied by the reviewers.
- Reviewers identify issues or problems and inform the author either verbally or in a documented form, which might be as formal as raising a defect report or as informal as annotating the document under review.
- The author decides on any action to take in response to the comments and updates the document accordingly.

Review Process 3/7

Work Product Review Process



Review Process 3/7

Phase 1/5 - Planning

- Defining the scope of the review
- Estimating effort & timeframe for the review
- Identifying review characteristics (review type, roles, checklists)
- Selecting reviewers & assigning roles
- Defining entry & exit criteria (for more formal review types)
- Checking that entry criteria are met before the review commences



Review Process 3/7

Phase 1/5 - Planning

Entry criteria

- Not many major defects found by a short check (by review leader)
- Line numbers (if relevant) present in the work product
- Clean-up work to remove format / syntax mild issues
- Stable and available related material
- Prepared & confident author

Exit criteria

- Average number of critical and/or major defects found per page
  - For example, no more than 3 critical and/or major defects found per page
- All pages have been checked at the right rate
- Defects fixed, deferred, etc

Review Process 3/7

Phase 2/5 - Initiate Review

- Distributing the work products & relevant material
- Explaining scope, objectives, process, roles & work products under review to the participants
- Answering any questions

Goal: get everyone on the same page



*Recorded **increase of 70% more major defects found per page** as a result of performing a kick-off meeting (a form of review initiation)*

Review Process 3/7

Phase 3/5 - Individual Review

- Reviewing all or part of the work document(s)
- Noting potential defects, recommendations & questions
- Use of checklists
- Checking rate (i.e., #pages reviewed/hr.)
  - Depends on type of work products, its complexity, # related work products, reviewer's experiences
  - Less formal reviews: 5 – 10 pages/hr.
  - Inspections: 1 page/hr.



Review Process 3/7

Phase 4/5 - Issue Comm. & Analysis

- Logging in a review meeting (severity levels: critical, major, minor)
  - No real discussion – focus on is logging as many defects as possible
- Discussion
  - More formal review: logging & discussion parts are separated
  - Less formal review: logging mixed with discussion
  - Moderator takes care of people issues & how discussed items are handled
- Decision marking: Evaluating the review findings against the exit criteria to make a review decision



Review Process 3/7

Phase 5/5 - Fixing & Reporting

- Creating defect reports
- Fixing defects (typically by author)
- Communicating defects to the appropriate person/team
- Recording updated status of defects as they are fixed
- Gathering metrics (for more formal review types; e.g., defects fixed, deferred, etc.)
- Checking that exit criteria are met
- Accepting work product when exit criteria is reached

Roles in Formal Review 1/2

- Author
  - Creates the work product under review
  - Fixes defects in the work product under review (if necessary)
- Management
  - Is responsible for review planning
  - Executes control decisions in the event of inadequate outcomes
  - Decides on the execution of reviews
  - Assigns staff, budget, and time
  - Monitors ongoing cost-effectiveness
- Facilitator/Moderator
  - Ensures effective running of review meetings (when held)
  - Mediates, if necessary, between the various points of view
  - Is often the person upon whom the success of the review depends

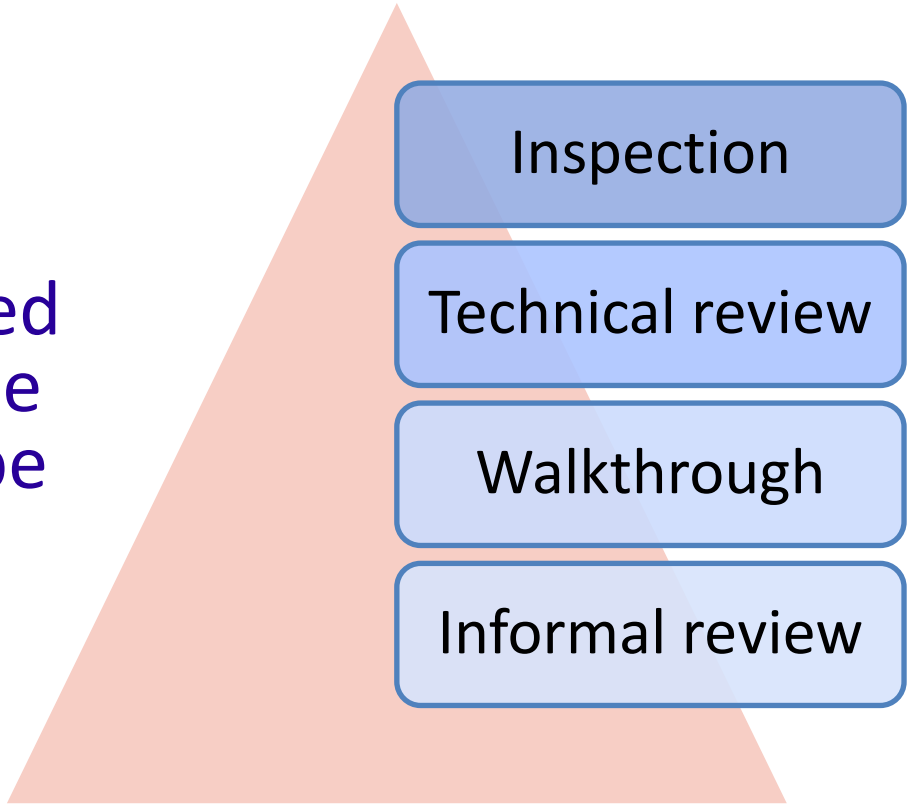
Roles in Formal Review 2/2

- Review Leader
  - Takes overall responsibility for the review
  - Decides review participants, time, and place
  - This role might be taken by a manager or the facilitator
- Reviewer
  - May be subject matter experts, persons working on the project, stakeholders with an interest in the work product, and/or individuals with specific technical or business backgrounds
  - Identify potential defects in the work product under review
  - May represent different perspectives
- Scribe (Recorder)
  - Recording new potential defects, open points and decisions from the review meetings (when held)

Review Process 5/7

Types of Review 1/2

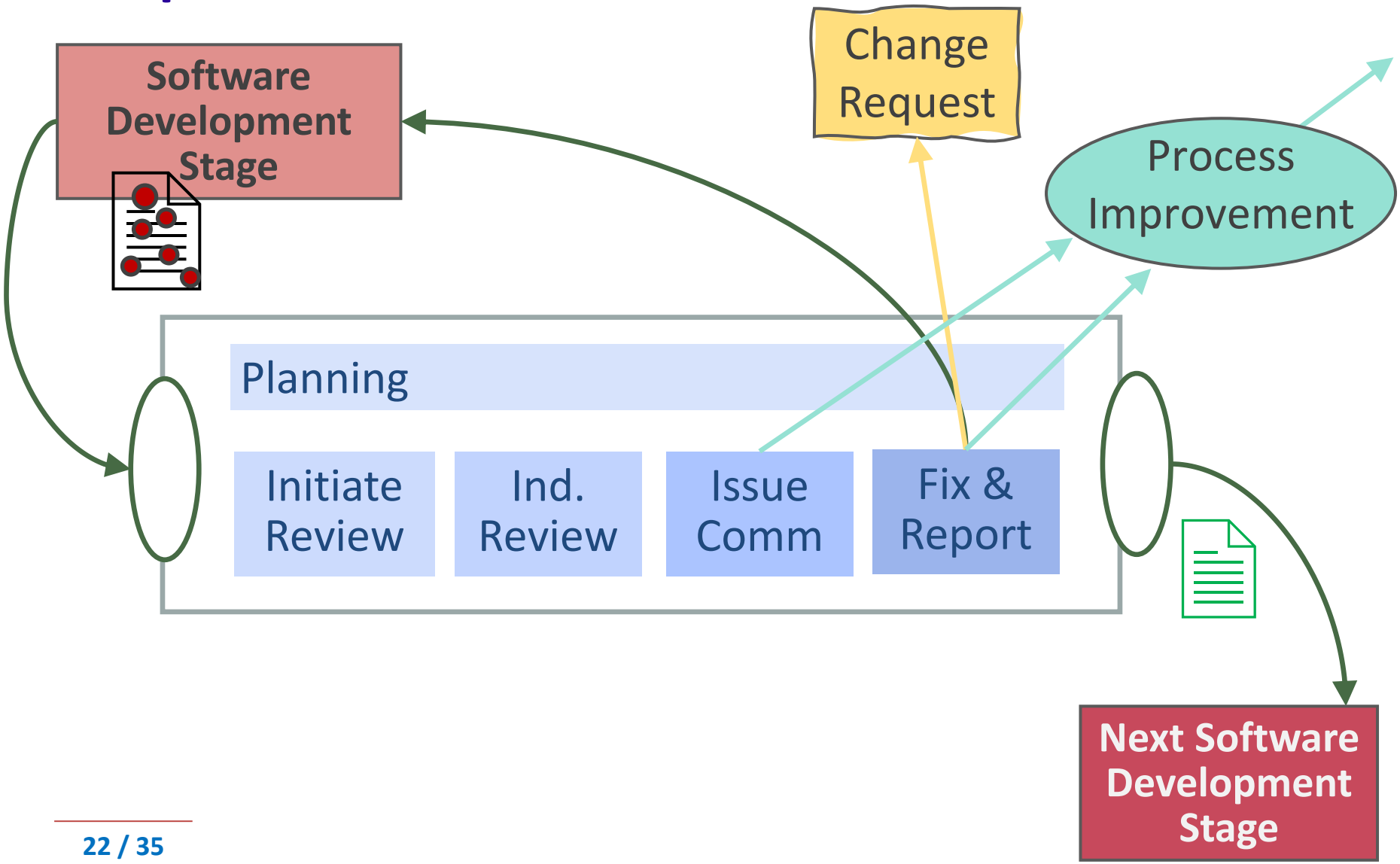
- Any work product should be reviewed.
- Type of review should be chosen based on a factors such as the need of the project, available resources, product type and risks, business domain, company culture, etc.
- One work product might go through more than one type of review.



Review Process 5/7

Types of Review 2/2

Inspection Process



Review Process 6/7

Review Techniques

- Ad hoc reviewing: carried out by independent reviewers informally, without a structured process
- Checklist-based reviewing: guided by a list of questions or required attributes
- Scenario-based reviewing: guided by determining the ability of the work product to address specific scenario
- Role-based reviewing: reviewers evaluate a work product from the perspective of different stakeholder roles
- Perspective-based reading/reviewing: reviewers evaluate the work product from the different viewpoints

Success factors for reviews

Organizational success factors for reviews

- Have clear objectives
- Pick the right review type & technique
- Review materials need to be kept up to date
- Limit the scope of the review
- It takes time
- Management support is critical

People-related success factors for reviews

- Pick the right reviewers
- Involve testers: to learn & to design test
- Each participant does their review work well
- Limit the scope of the review & pick things that really count
- Defects fond should be welcomed
- Review meetings are well managed
- Trust is critical
- How you communicated is important
- Follow the rules by keep it simple
- Train participants
- Continuously improve process & tools

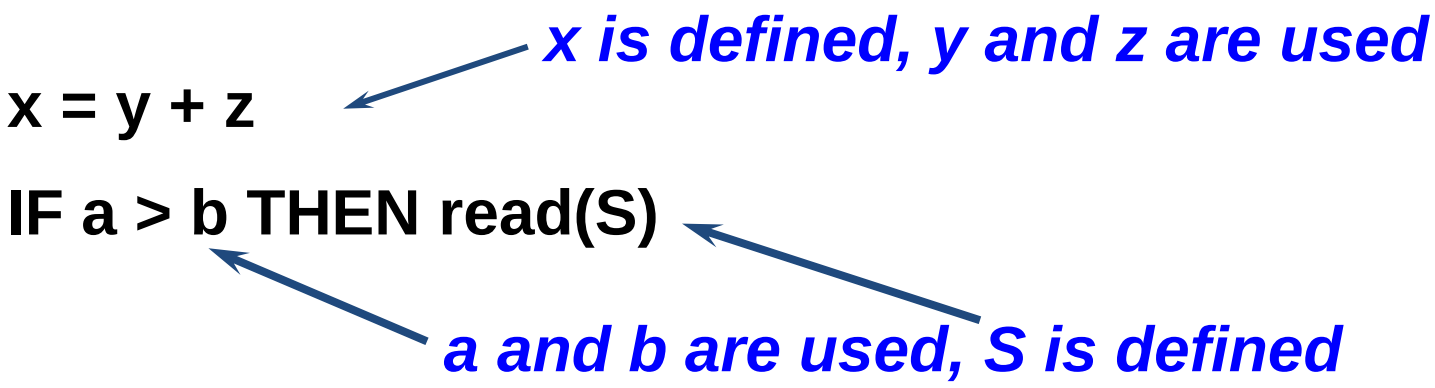
Static Analysis By Tools

- Static techniques do not execute the code
- A form of automated testing
  - check for violations of standards
  - check for things which may be a fault
- Descended from compiler technology
  - a compiler statically analyses code, and “knows” a lot about it, e.g. variable usage; finds syntax faults
  - static analysis tools extend this knowledge
  - can find unreachable code, undeclared variables, parameter type mis-matches, uncalled functions & procedures, array bound violations, etc.

Static Analysis By Tools

Data flow analysis

- This is the study of program variables
  - variable defined* where a value is stored into it
  - variable used where the stored value is accessed
  - variable is undefined before it is defined or when it goes out of scope



defined should not be confused with declared

Static Analysis By Tools

Data flow analysis faults

```
n := 0
read (x)
n := 1
while x > y do
    begin
      read (y)
      write( n*y)
      x := x - n
    end
```

Data flow abnomaly: n is re-defined without being used

Data flow fault: y is used before it has been defined (first time around the loop)

Static Analysis By Tools

Control flow analysis

- Highlights:
  - nodes not accessible from start node
  - infinite loops
  - multiple entry to loops
  - whether code is well structured, i.e. reducible
  - whether code conforms to a flowchart grammar
  - any jumps to undefined labels
  - any labels not jumped to
  - cyclomatic complexity and other metrics

Static Analysis By Tools
Control flow analysis

Unreachable code example

- Macro definitions
  (different for different platforms the code runs on)

```
Buffsize: 1000
Mailboxmax: 1000
IF Buffsize < Mailboxmax THEN
        Error-Exit
ENDIF
```

- Static Analysis finds the THEN clause unreachable, so will flag a fault

Cyclomatic complexity

- cyclomatic complexity is a measure of the complexity of a flow graph

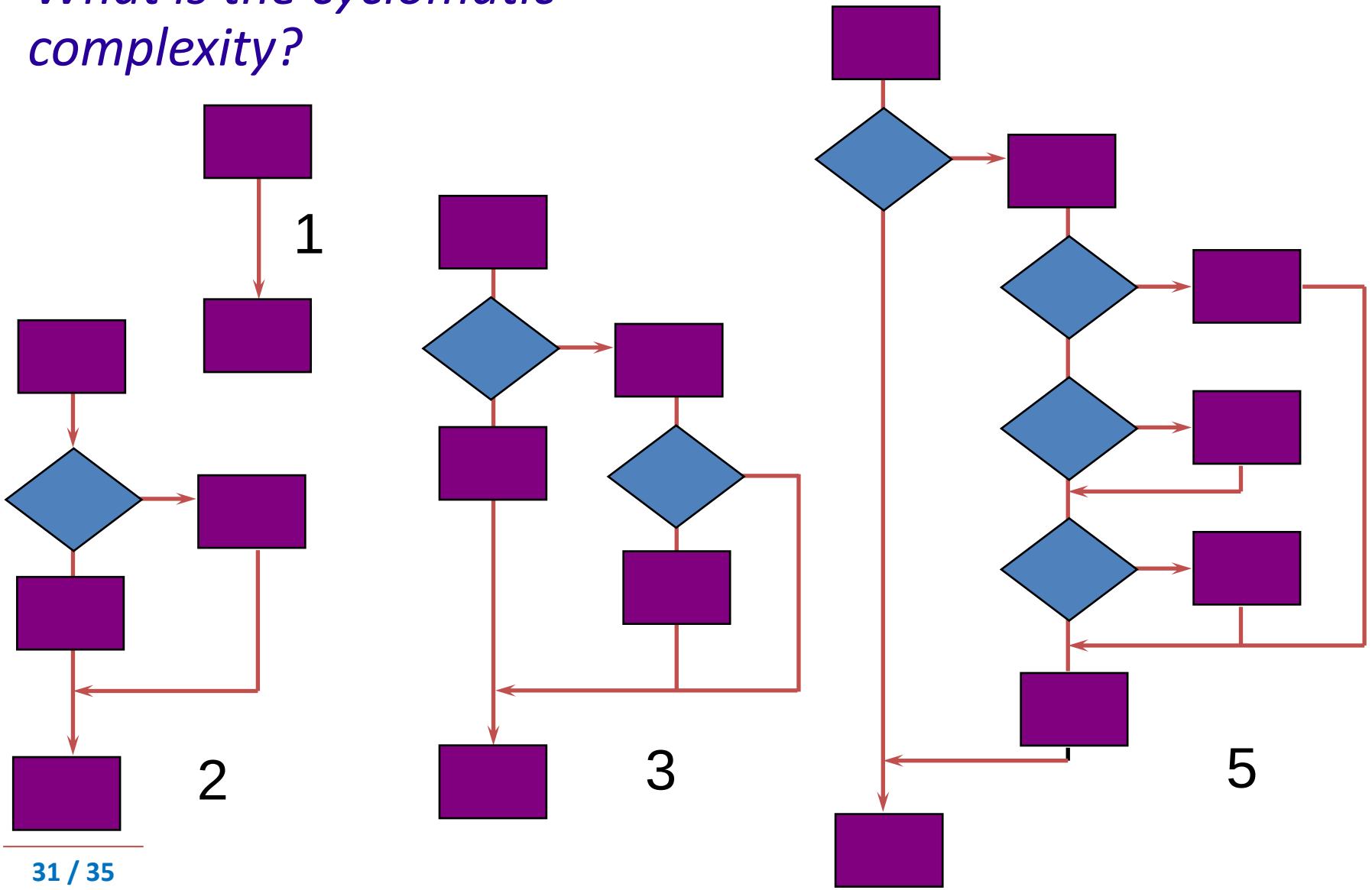
  (and therefore the code that the flow graph represents)
- the more complex the flow graph, the greater the measure
- it can most easily be calculated as:

  complexity = number of decisions + 1

Static Analysis By Tools

Control flow analysis

What is the cyclomatic complexity?



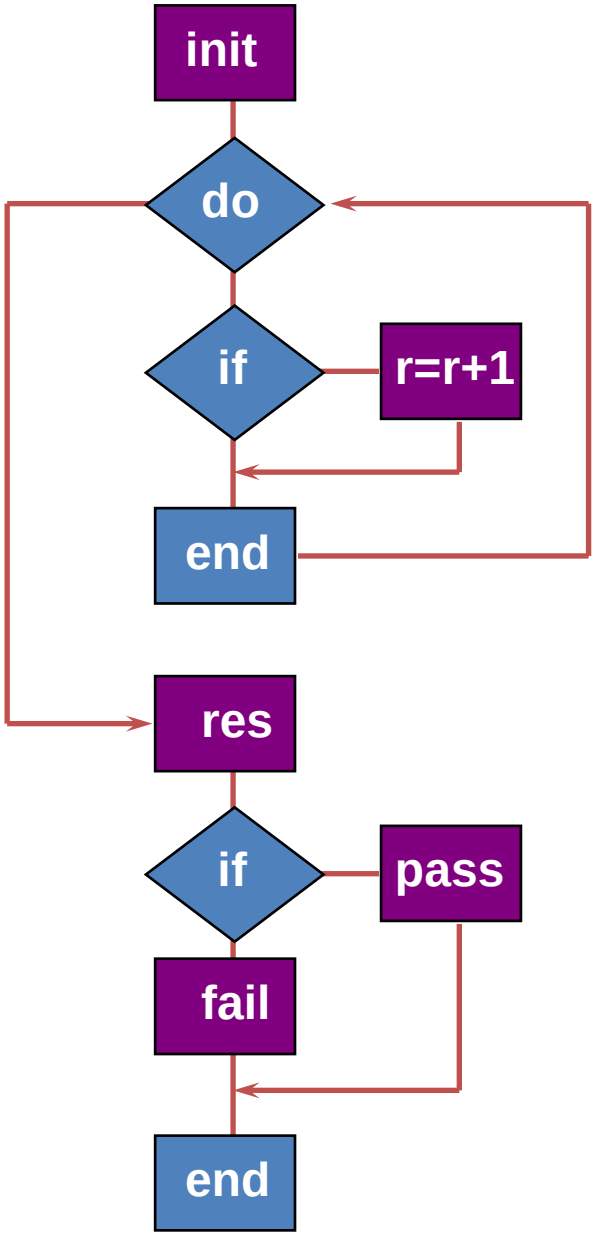
Static Analysis By Tools

Control flow analysis

Example control flow graph

Pseudo-code:

```
Result = 0
Right = 0
DO WHILE more Questions
   IF Answer = Correct THEN
        Right = Right + 1
   ENDIF
END DO
Result = (Right / Questions)
IF Result > 60% THEN
  Print "pass"
ELSE
  Print "fail”
ENDIF
```



Static Analysis By Tools

Static metrics

- lines of code (LOC)
- operands & operators (Halstead's metrics)
- fan-in & fan-out
- nesting levels
- function calls
- OO metrics:
  - inheritance tree depth,
  - number of methods,
  - coupling & cohesion

Static Analysis By Tools

Limitations and advantages

- Limitations:
  - cannot distinguish "fail-safe" code from programming faults or anomalies (often creates overload of spurious error messages)
  - does not execute the code, so not related to operating conditions
- Advantages:
  - can find faults difficult to "see"
  - gives objective quality assessment of code


Q
A